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Project Number: #2
Project Name: Functional Decomposition

Project Report #2

1. **What your own-choice quantity was and how did it fit into the simulation?**

   My own-choice quantity for the simulation is a Farmer() agent. The Farmers make decisions based on the current state of the simulation and take actions that affect the deer population and grain growth.

   In the simulation, the Farmer() agent has the following behaviors:

   > 1. If there is a 20% chance (1 in 5) determined randomly, the Farmer introduces a factor that affects 20% of the deer population which is shown below in the as NowDeerFarmer colum.
   >
   > 2. If there is a 10% chance (1 in 10) determined randomly, the Farmer will introduce a facetor that affects 30% of the grain growth.

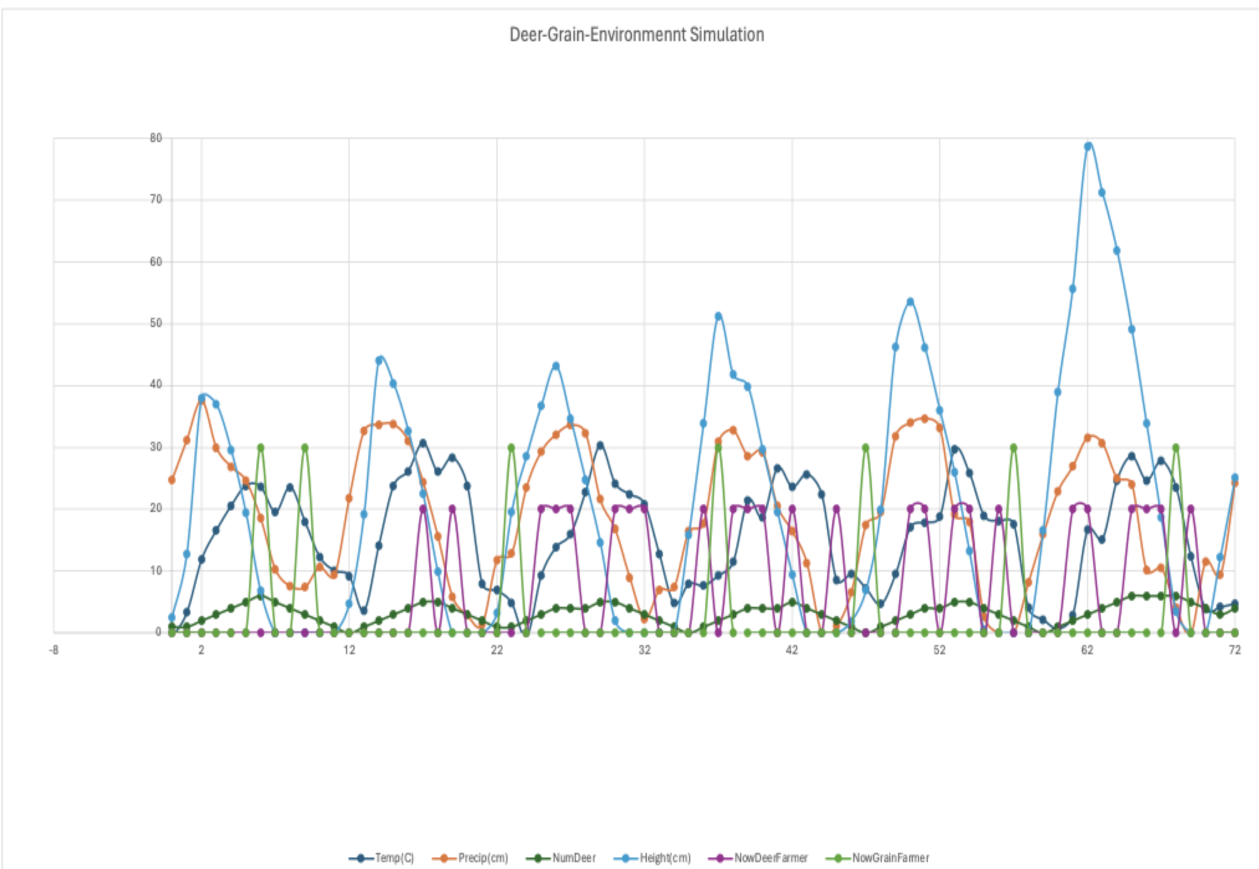
   These actions represent the Farmer's decisions to either affect the deer population by hunting or intervene in grain growth (by planting more grain or cutting down grain).

2. **A table showing values for temperature, precipitation, number of deer, height of the grain, and your own-choice quantity as a function of month number.**

There are 72 months, my month is from 0 -71, the last two columns(NowDeerFarmer,NowGrain Farmer) are my own-choice quantity.

Month	Temp(C)	Precip(cm)	NumDeer	Height(cm)	NowDeerFarmer	NowGrainFarmer
0	-0.73	24.77	1	2.54	0	0
1	3.33	31.23	1	12.81	0	0
2	11.88	37.58	2	38.04	0	0
3	16.59	29.97	3	37	0	0
4	20.56	26.88	4	29.63	0	0
5	23.82	24.62	5	19.48	0	0
6	23.61	18.55	6	6.78	0	30
7	19.54	10.28	5	0	0	0
8	23.57	7.61	4	0	0	0
9	18	7.42	3	0	0	30
10	12.28	10.69	2	0	0	0
11	10.08	9.36	1	0	0	0
12	9.2	21.82	0	4.77	0	0
13	3.65	32.65	1	19.13	0	0
14	14.12	33.71	2	44.11	0	0
15	23.8	33.73	3	40.35	0	0
16	26.17	31.08	4	32.73	0	0
17	30.73	24.4	5	22.57	20	0
18	26.06	15.65	5	9.87	0	0
19	28.38	5.83	4	0	20	0
20	23.75	2.86	3	0	0	0
21	7.96	1.33	2	0	0	0
22	6.89	11.79	1	3.24	0	0
23	4.83	12.86	1	19.56	0	30
24	-0.62	23.51	2	28.55	0	0
25	9.26	29.32	3	36.69	20	0
26	13.83	32.1	4	43.13	20	0
27	16.03	33.6	4	34.61	20	0
28	22.77	32.33	4	24.8	0	0
29	30.31	21.6	5	14.64	0	0
30	24.19	16.83	5	1.94	20	0
31	22.39	8.97	4	0	20	0
32	20.82	2.26	3	0	20	0
33	12.76	7	2	0	0	0
34	4.78	7.43	1	0	0	0
35	7.97	16.42	0	15.86	0	0
36	7.65	17.72	1	33.85	20	0
37	9.29	30.95	2	51.25	0	30
38	11.47	32.78	3	41.83	20	0
39	21.46	28.64	4	39.88	20	0
40	18.68	29.26	4	29.72	20	0
41	26.66	20.52	4	19.61	0	0
42	23.58	16.47	5	9.45	20	0
43	25.67	11.27	4	0	0	0
44	22.37	0	3	0	0	0
45	8.6	1.1	2	0	20	0
46	9.54	6.54	1	1.9	0	0
47	7.21	17.48	0	6.94	0	30
48	4.67	19.55	1	19.98	0	0
49	9.52	31.85	2	46.3	0	0
50	17.03	33.98	3	53.6	20	0
51	17.87	34.7	4	46.14	20	0
52	18.8	33.17	4	36.06	0	0
53	29.65	19.13	5	25.94	20	0
54	25.83	17.9	5	13.24	20	0
55	18.92	2.6	4	0.54	0	0
56	18.12	0	3	0	20	0
57	17.6	0	2	0	0	30
58	4.04	8.15	1	0	0	0
59	2.17	15.95	0	16.57	0	0
60	0.71	22.92	1	39	0	0
61	2.89	26.92	2	55.69	20	0
62	16.67	31.6	3	78.69	20	0
63	15.14	30.7	4	71.3	0	0
64	24.69	25.03	5	61.86	0	0
65	28.53	24.04	6	49.16	20	0
66	24.64	10.18	6	33.92	20	0
67	27.83	10.52	6	18.68	20	0
68	23.48	4.14	6	3.44	0	30
69	12.36	0	5	0	20	0
70	3.88	11.57	4	0	0	0
71	4.2	9.37	3	12.27	0	0

3. **A graph showing temperature, precipitation, number of deer, height of the grain, and your own-choice quantity as a function of month number. Note: if you change the units to °C and centimeters, the quantities might fit better on the same set of axes.**
cm = inches * 2.54
°C = (5./9.)*(°F-32)
This will make your heights have larger numbers and your temperatures have smaller numbers.



4. **A commentary about the patterns in the graph and why they turned out that way. What evidence in the curves proves that your own quantity is actually affecting the simulation correctly?**

   From the graph above, we observe a strong correlation among several parameters: temperature, precipitation, the number of deer, grain height, NowDeerFarmer, and NowGrainFarmer.

   For instance, when temperature(dark blue line) and precipitation(orange line), these two parameters show regular fluctuations, indicating seasonal changes, they likely have an indirect impact on the other parameters, such as the height of grain and the population of deers. The number of deers(dark green line) increase, likely stemming from favorable weather conditions promirting grain growth which, in tern, provide more food for deer, leading to a population increase. If we focusing on the "Number of Deer" as the primary quantity, its influence on the simulation becomes evident through its direct correlation with other parameters. Whenever temperature or precipitation increases (favorable conditions for food growth), there's a corresponding peak in the deer population, indicating food(grain) availability for their growth. This correlation confirms that the "Number of Deer" effectively impacts the simulation.

   Similarly, NowGrainFarmer (light green line) and NowDeerFarmer (purple line) exhibit a similar pattern as expected. This thread (Farmer()) introduces a random factor that affects both the deer population and grain growth. The factor is determined by a random number and can either influence 20% of the deer population or 30% of the grain growth. As shown in the graph above, these two lines randomly fluctuate with values of 20 or 30, as I configured in my agent.

   In conclusion, the patterns in the graph and the correlations between the different parameters provide evidence that my quantity is affecting the simulation correctly.